VSDBabySoC
Multi-Corner Static Timing Analysis

Assessment Date: 19/09/2026
Tool: OpenSTA 3.1.0
Design: VSDBabySoC
Technology: SKY130 HD Standard Cell Library

Objective

The objective of this analysis is to perform Static Timing Analysis (STA) of the synthesized VSDBabySoC design across all available SKY130 HD timing corners. Both minimum-delay (hold) and maximum-delay (setup) paths are analyzed.

1. STA Methodology

The synthesized Verilog netlist was analyzed using OpenSTA. The design constraints were loaded from the VSDBabySoC synthesis SDC file and each SKY130 HD Liberty timing corner was defined as a separate timing scene.

Detailed STA command used:

```
report_checks -scenes $corner -path_delay min_max -fields {net capacitance slew input_pin fanout}
```

The OpenSTA version used in this environment does not accept the literal field name **nets**. Therefore, the supported equivalent field names **net capacitance slew input_pin fanout** were used to generate the detailed reports.

Design files:

Synthesized netlist: output/synth/vsdbabysoc.synth.v
SDC constraints: src/sdc/vsdbabysoc_synthesis.sdc

2. PVT Corner Explanation

Notation	Meaning
FF	Fast NMOS / Fast PMOS process corner
SS	Slow NMOS / Slow PMOS process corner
TT	Typical NMOS / Typical PMOS process corner
n40C	Temperature = -40°C
025C	Temperature = 25°C
100C	Temperature = 100°C
1vXX	Nominal supply voltage in volts

For example, sky130_fd_sc_hd__ss_n40C_1v28 represents the slow-slow process corner at -40°C and a 1.28 V supply.

3. Multi-Corner STA Results

Corner	Hold Slack (ns)	Hold	Setup Slack (ns)	Setup
ff_100C_1v65	0.25	MET	4.05	MET
ff_100C_1v95	0.20	MET	5.57	MET
ff_n40C_1v56	0.29	MET	2.28	MET
ff_n40C_1v65	0.26	MET	3.41	MET
ff_n40C_1v76	0.22	MET	4.43	MET
ff_n40C_1v95	0.19	MET	5.59	MET
ss_100C_1v40	0.91	MET	-17.12	VIOLATED
ss_100C_1v60	0.64	MET	-7.87	VIOLATED
ss_n40C_1v28	1.83	MET	-62.56	VIOLATED
ss_n40C_1v35	1.35	MET	-39.70	VIOLATED
ss_n40C_1v40	1.12	MET	-29.69	VIOLATED
ss_n40C_1v44	0.99	MET	-23.91	VIOLATED
ss_n40C_1v60	0.66	MET	-10.61	VIOLATED
ss_n40C_1v76	0.50	MET	-4.38	VIOLATED
tt_025C_1v80	0.31	MET	1.94	MET
tt_100C_1v80	0.31	MET	2.09	MET

Observed result: All 16 timing corners have positive hold slack and therefore meet the reported hold requirement. Eight corners have positive setup slack, while the eight SS corners show negative setup slack and are reported as setup violations.

4. Detailed STA Output

The following sections provide the STA output for each of the 16 SKY130 HD timing corners. Each report contains both minimum-delay (hold) and maximum-delay (setup) analysis.

1. sky130_fd_sc_hd__ff_100C_1v65

Process: Fast-Fast (FF)
Temperature: 100°C
Supply Voltage: 1.65 V
Hold Slack: 0.25 ns (MET)
Setup Slack: 4.05 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_100C_1v65

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.22    0.22 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.22 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.22   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
---------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.22   data arrival time
---------------------------------------------------------------------------
                                  0.25   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_100C_1v65

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.04    0.00    0.24 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.09    0.09    0.34 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.09    0.00    0.34 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.07    0.08    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.07    0.00    0.42 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.56    0.42    0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.56    0.00    0.83 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.16    0.13    0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.16    0.00    0.96 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.19    1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.16    0.00    1.16 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.09    1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.07    0.00    1.25 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.15    1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.16    0.00    1.40 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.09    1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.07    0.00    1.49 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.15    1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.16    0.00    1.64 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.09    1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

2. sky130_fd_sc_hd__ff_100C_1v95

Process: Fast-Fast (FF)
Temperature: 100°C
Supply Voltage: 1.95 V
Hold Slack: 0.20 ns (MET)
Setup Slack: 5.57 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_100C_1v95

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.18    0.18 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.18 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.18   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.18   data arrival time
---------------------------------------------------------------------------
                                  0.20   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_100C_1v95

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.03    0.20    0.20 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.03    0.00    0.20 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.07    0.07    0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.07    0.00    0.27 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.06    0.07    0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.06    0.00    0.33 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.47    0.35    0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.47    0.00    0.68 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.14    0.09    0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.14    0.00    0.77 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.15    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.14    0.00    0.92 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.07    0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.07    0.00    0.99 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.12    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.14    0.00    1.11 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.07    1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.07    0.00    1.18 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.12    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.14    0.00    1.30 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.07    0.07    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

3. sky130_fd_sc_hd__ff_n40C_1v56

Process: Fast-Fast (FF)
Temperature: -40°C
Supply Voltage: 1.56 V
Hold Slack: 0.29 ns (MET)
Setup Slack: 2.28 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_n40C_1v56

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.26    0.26 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.26 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.26   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
---------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.26   data arrival time
---------------------------------------------------------------------------
                                  0.29   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_n40C_1v56

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.03    0.27    0.27 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.03    0.00    0.27 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.12    0.13    0.39 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.12    0.00    0.39 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.06    0.08    0.48 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.06    0.00    0.48 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    0.71    0.54    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.71    0.00    1.01 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.15    0.15    1.16 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.15    0.00    1.16 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.20    0.27    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.20    0.00    1.43 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.10    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.06    0.00    1.53 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.20    0.21    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.20    0.00    1.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.10    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.06    0.00    1.84 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.20    0.21    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.20    0.00    2.04 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.10    2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

4. sky130_fd_sc_hd__ff_n40C_1v65

Process: Fast-Fast (FF)
Temperature: -40°C
Supply Voltage: 1.65 V
Hold Slack: 0.26 ns (MET)
Setup Slack: 3.41 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_n40C_1v65

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.23    0.23 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.23 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.23   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.23   data arrival time
---------------------------------------------------------------------------
                                  0.26   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_n40C_1v65

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.03    0.24    0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.03    0.00    0.24 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.10    0.11    0.35 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.10    0.00    0.35 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.06    0.07    0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.06    0.00    0.42 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    0.64    0.48    0.90 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.64    0.00    0.90 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.14    0.12    1.02 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.14    0.00    1.02 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.18    0.23    1.26 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.18    0.00    1.26 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.09    1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.06    0.00    1.34 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.18    0.18    1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.18    0.00    1.53 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.09    1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.06    0.00    1.61 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.18    0.18    1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.18    0.00    1.79 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.09    1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

5. sky130_fd_sc_hd__ff_n40C_1v76

Process: Fast-Fast (FF)
Temperature: -40°C
Supply Voltage: 1.76 V
Hold Slack: 0.22 ns (MET)
Setup Slack: 4.43 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_n40C_1v76

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.20    0.20 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.20 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.20   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.20   data arrival time
---------------------------------------------------------------------------
                                  0.22   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[1][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_n40C_1v76

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.03    0.21    0.21 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.03    0.00    0.21 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.09    0.10    0.31 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.09    0.00    0.31 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.05    0.06    0.37 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.05    0.00    0.37 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    0.58    0.43    0.81 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.58    0.00    0.81 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.13    0.09    0.90 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.13    0.00    0.90 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.20    1.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.16    0.00    1.10 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.07    1.17 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.06    0.00    1.17 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.16    1.33 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.16    0.00    1.33 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.07    1.41 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.06    0.00    1.41 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.16    0.16    1.56 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.16    0.00    1.56 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.06    0.07    1.64 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

6. sky130_fd_sc_hd__ff_n40C_1v95

Process: Fast-Fast (FF)
Temperature: -40°C
Supply Voltage: 1.95 V
Hold Slack: 0.19 ns (MET)
Setup Slack: 5.59 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ff_n40C_1v95

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.02    0.17    0.17 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.02    0.00    0.17 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.17   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.17   data arrival time
---------------------------------------------------------------------------
                                  0.19   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ff_n40C_1v95

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.03    0.18    0.18 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.03    0.00    0.18 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.08    0.08    0.26 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.08    0.00    0.26 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.05    0.05    0.32 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.05    0.00    0.32 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.50    0.37    0.69 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.50    0.00    0.69 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.12    0.06    0.75 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.12    0.00    0.75 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.17    0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.14    0.00    0.92 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.05    0.06    0.98 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.05    0.00    0.98 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.13    1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.14    0.00    1.11 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.05    0.06    1.17 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.05    0.00    1.17 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.14    0.13    1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.14    0.00    1.30 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.05    0.06    1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

7. sky130_fd_sc_hd__ss_100C_1v40

Process: Slow-Slow (SS)
Temperature: 100°C
Supply Voltage: 1.40 V
Hold Slack: 0.91 ns (MET)
Setup Slack: -17.12 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_100C_1v40

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.75    0.75 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.05    0.00    0.75 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.75   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.15   -0.15   library hold time
                                 -0.15   data required time
---------------------------------------------------------------------------
                                 -0.15   data required time
                                 -0.75   data arrival time
---------------------------------------------------------------------------
                                  0.91   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_100C_1v40

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.14    0.91    0.91 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.14    0.00    0.91 ^ core/U54/A (sky130_fd_sc_hd__clkinv_1)
     2    0.01    0.15    0.21    1.13 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
                                         core/n145 (net)
                  0.15    0.00    1.13 v core/U477/A (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.13    0.18    1.30 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n146 (net)
                  0.13    0.00    1.30 ^ core/U51/B (sky130_fd_sc_hd__nand2_2)
    32    0.07    0.80    0.71    2.01 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
                                         core/n206 (net)
                  0.80    0.00    2.01 v core/U478/B1 (sky130_fd_sc_hd__a22o_1)
     1    0.00    0.12    0.80    2.81 v core/U478/X (sky130_fd_sc_hd__a22o_1)
                                         core/n148 (net)
                  0.12    0.00    2.81 v core/U479/A (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.36    0.47    3.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.36    0.00    3.28 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.41    0.58    3.86 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.41    0.00    3.86 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.21    0.36    4.22 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.21    0.00    4.22 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.41    0.48    4.70 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.41    0.00    4.70 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.21    0.36    5.05 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.21    0.00    5.05 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.41    0.48    5.53 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
```

8. sky130_fd_sc_hd__ss_100C_1v60

Process: Slow-Slow (SS)
Temperature: 100°C
Supply Voltage: 1.60 V
Hold Slack: 0.64 ns (MET)
Setup Slack: -7.87 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_100C_1v60

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.04    0.53    0.53 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.04    0.00    0.53 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.53   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.11   -0.11   library hold time
                                 -0.11   data required time
-----------------------------------------------------------------------------
                                 -0.11   data required time
                                 -0.53   data arrival time
-----------------------------------------------------------------------------
                                  0.64   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_100C_1v60

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.11    0.62    0.62 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.11    0.00    0.62 ^ core/U54/A (sky130_fd_sc_hd__clkinv_1)
     2    0.01    0.12    0.15    0.78 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
                                         core/n145 (net)
                  0.12    0.00    0.78 v core/U477/A (sky130_fd_sc_hd__nand2_1)
     1    0.00    0.10    0.13    0.91 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n146 (net)
                  0.10    0.00    0.91 ^ core/U51/B (sky130_fd_sc_hd__nand2_2)
    32    0.07    0.60    0.51    1.42 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
                                         core/n206 (net)
                  0.60    0.00    1.42 v core/U478/B1 (sky130_fd_sc_hd__a22o_1)
     1    0.00    0.09    0.56    1.98 v core/U478/X (sky130_fd_sc_hd__a22o_1)
                                         core/n148 (net)
                  0.09    0.00    1.98 v core/U479/A (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.29    0.33    2.31 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.29    0.00    2.31 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.31    0.42    2.72 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.31    0.00    2.72 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.24    2.97 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.16    0.00    2.97 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.31    0.34    3.31 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.31    0.00    3.31 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.16    0.24    3.55 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.16    0.00    3.55 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.31    0.34    3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
```

9. sky130_fd_sc_hd__ss_n40C_1v28

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.28 V
Hold Slack: 1.83 ns (MET)
Setup Slack: -62.56 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v28

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.07    1.45    1.45 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.07    0.00    1.45 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.45   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.38   -0.38   library hold time
                                 -0.38   data required time
---------------------------------------------------------------------------
                                 -0.38   data required time
                                 -1.45   data arrival time
---------------------------------------------------------------------------
                                  1.83   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v28

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.11    1.50    1.50 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.11    0.00    1.50 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.65    0.71    2.20 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.65    0.00    2.20 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.23    0.56    2.76 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.23    0.00    2.76 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.13    3.38    2.67    5.43 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  3.38    0.00    5.43 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.53    1.71    7.14 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.53    0.00    7.14 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    1.07    1.42    8.57 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  1.07    0.00    8.57 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.22    0.76    9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.22    0.00    9.33 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    1.07    1.20   10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  1.07    0.00   10.53 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.22    0.76   11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.22    0.00   11.29 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    1.07    1.20   12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  1.07    0.00   12.49 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.22    0.76   13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

10. sky130_fd_sc_hd__ss_n40C_1v35

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.35 V
Hold Slack: 1.35 ns (MET)
Setup Slack: -39.70 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v35

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.06    1.08    1.08 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.06    0.00    1.08 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.08   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.27   -0.27   library hold time
                                 -0.27   data required time
---------------------------------------------------------------------------
                                 -0.27   data required time
                                 -1.08   data arrival time
---------------------------------------------------------------------------
                                  1.35   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v35

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.09    1.12    1.12 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.09    0.00    1.12 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.47    0.52    1.64 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.47    0.00    1.64 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.19    0.41    2.05 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.19    0.00    2.05 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.13    2.53    2.01    4.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  2.53    0.00    4.06 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.41    1.22    5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.41    0.00    5.28 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.77    1.04    6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.77    0.00    6.33 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.17    0.54    6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.17    0.00    6.87 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.77    0.87    7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.77    0.00    7.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.17    0.54    8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.17    0.00    8.28 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.77    0.87    9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.77    0.00    9.15 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.17    0.54    9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

11. sky130_fd_sc_hd__ss_n40C_1v40

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.40 V
Hold Slack: 1.12 ns (MET)
Setup Slack: -29.69 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v40

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.91    0.91 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.05    0.00    0.91 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.91   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.22   -0.22   library hold time
                                 -0.22   data required time
---------------------------------------------------------------------------
                                 -0.22   data required time
                                 -0.91   data arrival time
---------------------------------------------------------------------------
                                  1.12   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v40

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.08    0.94    0.94 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.08    0.00    0.94 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.39    0.43    1.38 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.39    0.00    1.38 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.17    0.34    1.72 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.17    0.00    1.72 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    2.13    1.70    3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  2.13    0.00    3.42 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.36    0.99    4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.36    0.00    4.40 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.64    0.87    5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.64    0.00    5.27 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.15    0.44    5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.15    0.00    5.71 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.64    0.71    6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.64    0.00    6.43 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.15    0.44    6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.15    0.00    6.87 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.64    0.71    7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.64    0.00    7.58 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.15    0.44    8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

12. sky130_fd_sc_hd__ss_n40C_1v44

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.44 V
Hold Slack: 0.99 ns (MET)
Setup Slack: -23.91 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v44

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.05    0.80    0.80 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.05    0.00    0.80 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.80   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.19   -0.19   library hold time
                                 -0.19   data required time
---------------------------------------------------------------------------
                                 -0.19   data required time
                                 -0.80   data arrival time
---------------------------------------------------------------------------
                                  0.99   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v44

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.07    0.83    0.83 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.07    0.00    0.83 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.34    0.38    1.21 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.34    0.00    1.21 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.15    0.30    1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.15    0.00    1.51 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    1.89    1.50    3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  1.89    0.00    3.01 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.34    0.88    3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.34    0.00    3.88 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.56    0.77    4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.56    0.00    4.65 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.14    0.39    5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.14    0.00    5.04 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.56    0.62    5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.56    0.00    5.67 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.14    0.39    6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.14    0.00    6.06 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.56    0.62    6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.56    0.00    6.68 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.14    0.39    7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

13. sky130_fd_sc_hd__ss_n40C_1v60

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.60 V
Hold Slack: 0.66 ns (MET)
Setup Slack: -10.61 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v60

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.54    0.54 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.54 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.54   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.12   -0.12   library hold time
                                 -0.12   data required time
---------------------------------------------------------------------------
                                 -0.12   data required time
                                 -0.54   data arrival time
---------------------------------------------------------------------------
                                  0.66   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v60

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.05    0.57    0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.05    0.00    0.57 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.23    0.25    0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.23    0.00    0.82 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.12    0.20    1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.12    0.00    1.02 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.14    1.32    1.04    2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  1.32    0.00    2.06 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.25    0.53    2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.25    0.00    2.58 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.38    0.52    3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.38    0.00    3.10 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.11    0.24    3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.11    0.00    3.35 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.38    0.41    3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.38    0.00    3.76 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.11    0.24    4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.11    0.00    4.00 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.38    0.41    4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.38    0.00    4.41 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.11    0.24    4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

14. sky130_fd_sc_hd__ss_n40C_1v76

Process: Slow-Slow (SS)
Temperature: -40°C
Supply Voltage: 1.76 V
Hold Slack: 0.50 ns (MET)
Setup Slack: -4.38 ns (VIOLATED)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__ss_n40C_1v76

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.41    0.41 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.41 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.41   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.09   -0.09   library hold time
                                 -0.09   data required time
---------------------------------------------------------------------------
                                 -0.09   data required time
                                 -0.41   data arrival time
---------------------------------------------------------------------------
                                  0.50   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__ss_n40C_1v76

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.00    0.05    0.43    0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.05    0.00    0.43 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.17    0.19    0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.17    0.00    0.62 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.10    0.15    0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.10    0.00    0.77 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    1.02    0.80    1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  1.02    0.00    1.57 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.21    0.36    1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.21    0.00    1.93 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.29    0.39    2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.29    0.00    2.32 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.18    2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.09    0.00    2.50 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.29    0.31    2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.29    0.00    2.80 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.18    2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.09    0.00    2.98 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.29    0.31    3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.29    0.00    3.29 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.18    3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

15. sky130_fd_sc_hd__tt_025C_1v80

Process: Typical-Typical (TT)
Temperature: 25°C
Supply Voltage: 1.80 V
Hold Slack: 0.31 ns (MET)
Setup Slack: 1.94 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__tt_025C_1v80

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.27    0.27 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.27 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.27   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
---------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.27   data arrival time
---------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__tt_025C_1v80

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.28    0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.04    0.00    0.28 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.11    0.12    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.11    0.00    0.40 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.08    0.10    0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.08    0.00    0.50 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.67    0.51    1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.67    0.00    1.01 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.17    0.18    1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.17    0.00    1.19 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.25    1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.19    0.00    1.43 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.08    0.00    1.54 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.19    0.00    1.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.08    0.00    1.84 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.19    2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.19    0.00    2.04 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.08    0.11    2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

16. sky130_fd_sc_hd__tt_100C_1v80

Process: Typical-Typical (TT)
Temperature: 100°C
Supply Voltage: 1.80 V
Hold Slack: 0.31 ns (MET)
Setup Slack: 2.09 ns (MET)

STA output excerpt:

```
Startpoint: core/CPU_imm_a2_reg[10]
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_imm_a3_reg[10]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Corner: sky130_fd_sc_hd__tt_100C_1v80

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
     1    0.00    0.03    0.28    0.28 ^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_imm_a2[10] (net)
                  0.03    0.00    0.28 ^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.28   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04   -0.04   library hold time
                                 -0.04   data required time
---------------------------------------------------------------------------
                                 -0.04   data required time
                                 -0.28   data arrival time
---------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: core/CPU_is_addi_a3_reg
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: sky130_fd_sc_hd__tt_100C_1v80

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
     2    0.01    0.04    0.29    0.29 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
                                         core/CPU_is_addi_a3 (net)
                  0.04    0.00    0.29 v core/U474/A (sky130_fd_sc_hd__nor2_1)
     2    0.00    0.11    0.11    0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
                                         core/n144 (net)
                  0.11    0.00    0.40 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
     2    0.01    0.09    0.10    0.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
                                         core/n147 (net)
                  0.09    0.00    0.51 v core/U9/A (sky130_fd_sc_hd__inv_2)
    34    0.15    0.64    0.48    0.99 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
                                         core/n150 (net)
                  0.64    0.00    0.99 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
     2    0.01    0.18    0.19    1.18 v core/U479/X (sky130_fd_sc_hd__xor2_1)
                                         core/n210 (net)
                  0.18    0.00    1.18 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.23    1.41 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n809 (net)
                  0.19    0.00    1.41 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.12    1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n791 (net)
                  0.09    0.00    1.53 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.18    1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n774 (net)
                  0.19    0.00    1.71 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.11    1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
                                         core/n755 (net)
                  0.09    0.00    1.83 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
     2    0.01    0.19    0.18    2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
                                         core/n740 (net)
                  0.19    0.00    2.01 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
     2    0.01    0.09    0.11    2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
```

5. Conclusion

Multi-corner Static Timing Analysis was completed for all 16 available SKY130 HD timing corners using OpenSTA 3.1.0. Both minimum-delay (hold) and maximum-delay (setup) checks were evaluated across the specified PVT conditions.

The hold analysis is clean across all 16 corners, with positive hold slack in every case. Setup timing, however, is not closed across all corners: 8 corners meet the setup requirement with positive slack, while all 8 SS (slow-slow) corners report setup violations.

The most critical setup result is at ss_n40C_1v28, with a setup slack of -62.56 ns. The negative setup slack becomes less severe as the supply voltage increases, but setup violations remain at the SS corners, including -4.38 ns at ss_n40C_1v76.

Therefore, the synthesized design satisfies the reported hold requirements but requires further setup-timing optimization for the slow-slow PVT conditions before it can be considered timing-clean across all analyzed corners. The detailed corner reports are stored in output/sta/corners/.